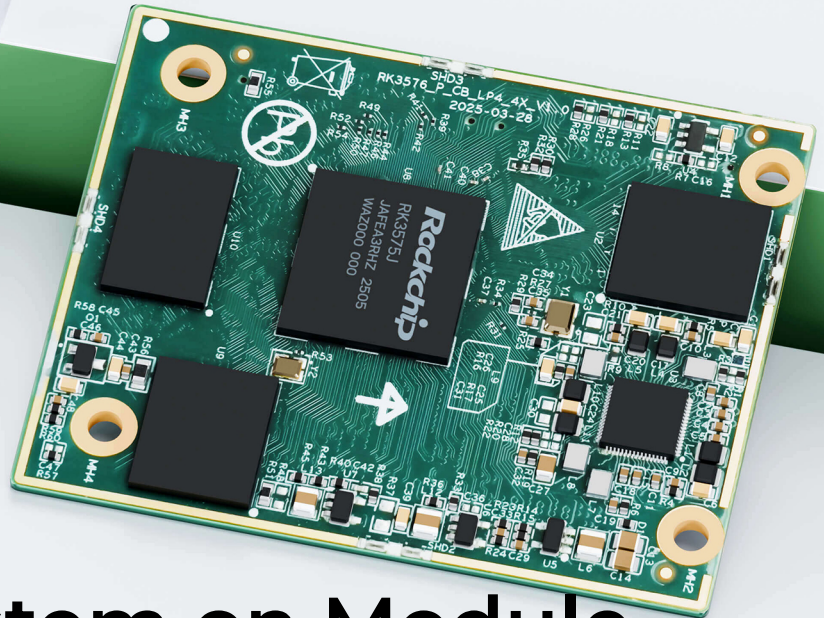
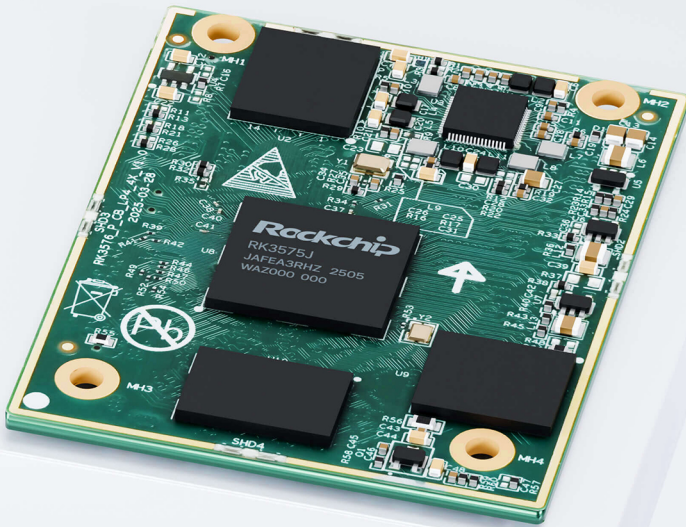


Embrace Edge AI, Empower Industrial Digitalization



EC3576-C AI System on Module

· RK3576J ARM SoC

· 4×A72+4×A53

· 6TOPS NPU

· 8K Video Codec

1. Product Overview

EC3576-C system on module is powered by the Rockchip RK3576 processor, tailored for AIoT and industrial markets.

Product Features:

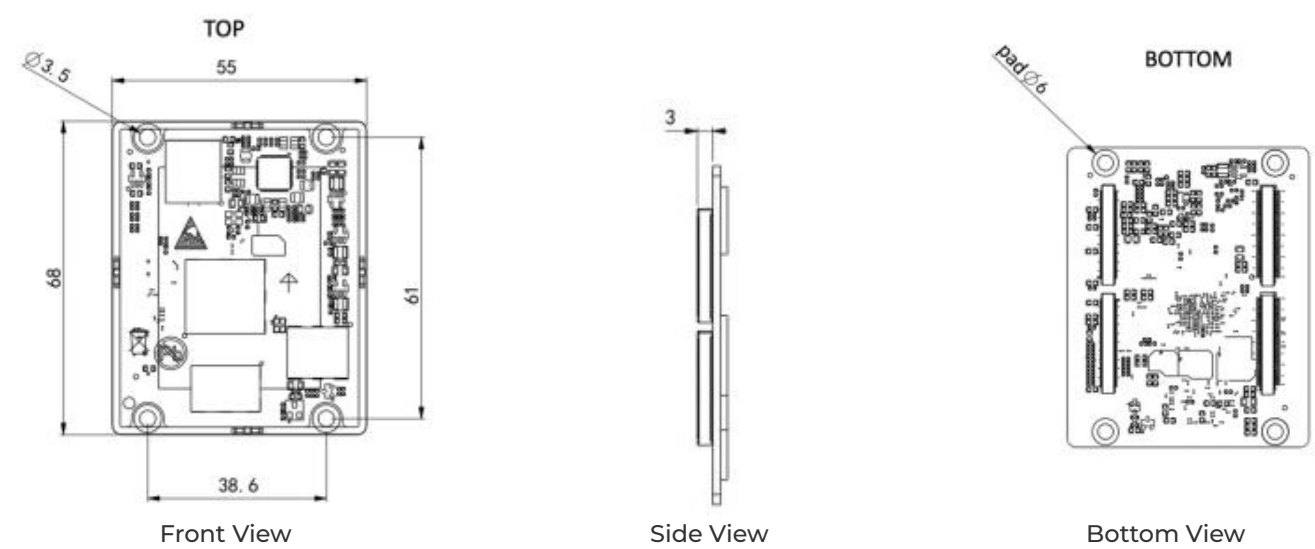
- **High-Performance CPU:** 8nm octa-core, 4×A72+4×A53 @ 2.3 GHz
- **Powerful AI NPU:** 6.0 TOPS INT8, multi-format support
- **Advanced Graphics:** Mali-G52 MC3, 8K codec, triple display
- **Rich Interfaces:** PCIe, USB 3.2, CAN-FD, GMAC, MIPI
- **Industrial Grade:** -40 °C ~ +85 °C, ultra-thin connector

Core Technical Specifications

Technical Indicator	Specification
OS	Android 14
Industrial Protocol	Modbus RTU/TCP, EtherNet/IP, OPC UA, Mitsubishi MC
Remote Management	InHand DeviceLive, HTTPS, SSH
AI	6.0 TOPS INT8 NPU
Video Codec	8K@30 fps decode, 4K@60 fps encode
Cloud Platform	Cloud parameter config, container management, firmware management
CPU	4×A72 @ 2.3 GHz + 4×A53 @ 2.2 GHz
Memory / Storage	8 GB LPDDR4 / 256 GB UFS
Display Interface	HDMI/eDP/DSI/DP, triple independent display
Connectivity	2×GMAC, USB 3.2, CAN-FD, MIPI-CSI×5, UART×12

Technical Indicator	Specification
Dimensions (W × D × H)	55 × 68 × 3 mm
Operating Temperature	-40 °C ~ +85 °C

2. Product Dimensions



Note:

- 1. All dimensions are in millimeters (mm).
- 2. All dimensions are approximate and for reference only.
- 3. Dimensioned drawings are not intended for machining.
- 4. Dimensions are subject to part and manufacturing tolerances.
- 5. Specifications may change without prior notice.

3. Hardware Specifications

Category/Parameter	Specification
Hardware Platform	

Category/Parameter	Specification
CPU	4 × ARM Cortex-A72 @ 2.3 GHz + 4 × ARM Cortex-A53 @ 2.2 GHz
NPU	6.0 TOPS INT8, supports INT4/INT8/INT16/FP16/BF16/TF32
GPU	ARM Mali-G52 MC3
VPU	Hard Encode: H.264, H.265, 4K@60 fps; Hard Decode: H.264, H.265, VP9, AV1, AVS2, 8K@30 fps or 4K@120 fps
RAM	8 GB LPDDR4
ROM	256 GB UFS
Display Interface	
HDMI/eDP TX	1 × HDMI v2.1 / eDP v1.3 (muxed), up to 4K@120 Hz; Supports CEC, ARC, HDCP v2.3/v1.4; eDP supports up to 3 displays with different content
MIPI DSI	1 × MIPI DSI-2 TX, D-PHY v2.0 (4 lanes) or C-PHY v1.1 (3 trios), up to 2560 × 1600@60 Hz
Parallel	1 × Parallel RGB/BT.656/BT.1120, up to 1920 × 1080@60 Hz
EBC	1 × E-ink EPD, 2560 × 1920 hard decoding, 16-bit data bus, up to 32-level grayscale
DP TX	1 × USB/DP combo, DisplayPort v1.4, up to 4K@120 Hz, supports MST, USB Type-C with DP Alt, HDCP v2.3/v1.3
Camera Interface	
MIPI-CSI	5 × CSI-2; 4 × 2-lane D-PHY v1.2 (2.5 Gbps/lane, combinable to 2 × 4-lane); 1 × 4-lane D-PHY v2.0 (4.5 Gbps/lane) or 3 C-PHY trios; Supports up to 5 cameras working together
DVP	1 × 8/10/12/16-bit standard DVP, up to 150 MHz data input; Supports BT.601/BT.656/BT.1120

Category/Parameter	Specification
Audio	
SAI	≤5; SAI 0/1 support 4 TX lanes + 4 RX lanes; SAI 2/3/4 support 1 TX lane + 1 RX lane; Supports I2S/TDM/PCM mode, sampling rate up to 192 kHz, 16~32 bits
Digital Audio Codec	1; Supports 2 × DAC, 3 blending modes each; Supports I2S/PCM master/slave mode, 16-bit sampling rate; Supports volume control
Connectivity	
Ethernet	≤2 × GMAC, pinout by RGMII/RMII, 10/100/1000 Mbps
SDIO	≤2, SDIO v3.0, 4-bit
USB 3.2	2 (1 × Type-C supports DP Alt mode; 1 × combo high speed interface)
USB 2.0 OTG	2
UART	≤12, 64-bit FIFO for TX/RX; Supports 5/6/7/8-bit serial transceiving, up to 4 Mbps; All 12 UARTs support flow control and RS485
CAN-FD	≤2, compliant with CAN and CAN-FD; Supports standard and extended frames; 8192-bit FIFO
SPI	≤5, master and slave mode, each supports two chip selections
I2C	≤9, 7-bit and 10-bit address modes; Standard mode 100 kbps, HS mode 400 kbps
PWM	≤16, supports interrupt operation, capturing mode
Power	
Input Power	DC 5V

Category/Parameter	Specification
Mechanical	
Dimensions (W × D × H)	55 × 68 × 3 mm
Package	Board-to-board connector (4 × 100-pin, 0.4 mm pitch, combined height 1.5 mm)
Mounting Holes	4 × φ3.5 mm
Environment	
Operating Temperature	-40 °C ~ +85 °C

4. Software Specifications

Category/Parameter	Specification
Operating System	
OS	Android 14
OS Flashing Method	USB OTG
Data Acquisition Protocol (DSA)	
Industrial Protocol	Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINS UDP, Host Link, PPI
Electricity Protocol	DLT645-2007, IEC101/104, DNP3.0
Other Protocol	BACnet, CNC

Category/Parameter	Specification
Maintenance and Management	
Upgrade Method	Supports patent upgrade mechanism, local or remote firmware upgrade
Log	Supports local system logs, remote logs, important log power-off preservation
Remote Management	InHand DeviceLive, HTTP, HTTPS, SSH, etc.
DeviceLive Cloud	Supports cloud-based parameter configuration, container management, application and firmware management

5. Ordering Information

Model Code

Model code: EC3576-C

Product Model

Model	CPU	CPU Clock Speed	RAM	ROM	Operating Temperature
EC3576-C	4 × A72 + 4 × A53	A7 @ 2.3 GHz, A53 @ 2.2 GHz	8 GB	256 GB	-40 °C ~ +85 °C

6. Contact Us

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